

# Interactive Weather Dashboard

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## Introduction:

Dashboard is a simple graphical presentation and summary of data that has been analyzed for a particular purpose, with consideration of details which helps to easier understanding and conclusions. Such as (Michael & John, 2013, p. 12), "A dashboard is a visual interface that provides at-a-glance views into key measures relevant to a particular objective or business process". One of the features of dashboard is that it presents only the relevant data with the goal and keeping the reader focused on the outcome and away from additional analysis. Therefore, the main goal of this dashboard is to analyze daily and monthly weather conditions and trends for informed decision-making. According to Michael and John's (2013) dashboard design principle, the key features such as "Keep it simple, Use the right titles and labels, Format the numbers effectively and Remove unnecessary charts," have been identified and applied to the effective design of this dashboard.

## Dashboard overview:

### Main goals:

The main goal of the dashboard is to analyze and visualize the daily and monthly weather conditions and trends for informed decision-making.

Understanding user requirements helps design an effective dashboard that presents the main objective with precise details. Michael and John (2013), argue that collecting user requirements is a fundamental step in the development of a dashboard and should consider the data sources, data types and data

dimensions. Accordingly considering all the user requirements and data sources for designing this dashboard, the main goals of this dashboard are to track weather conditions on a daily and monthly basis, with details such as (highest and lowest temperatures and rainfall amount on a particular day in different location) also to display a summary of the situation for each month optionally in different locations.

### Dashboard layout:

Combination of three simple and user-friendly frameworks (Daily outlook, Monthly outlook, and monthly liner graph) provides the details to help with better decision making.

### Daily outlook:

The daily outlook section contains the Highest and lowest temperatures, as well as the amount of rainfall for a specific date in two different locations: Kabul city and Mazar-e-Sharif. Because of flexibility using the drop-down list allowing user to choose a specific date. Also, condition formatting and basic functions like: VLOOKUP, MIN, MAX and AVG used to extract and analyze the outputs.

| <div>  <b>Daily outlook</b> </div> |           |        |                                                                                       |
|-----------------------------------------------------------------------------------------------------------------------|-----------|--------|---------------------------------------------------------------------------------------|
| select date                                                                                                           | 5/01/2024 | Kabul  | Mazar-e-Sharif                                                                        |
| Rainfall Mount                                                                                                        | 0.0 mm    | 0.2 mm |  |
| Highest                                                                                                               | 35.0 C    | 36.3 C |  |
| Lowest                                                                                                                | 21.6 C    | 21.7 C |  |

Figure 1: Daily outlook, weather dashboard

## Monthly outlook:

One of the goals of the dashboard is to display weather conditions on monthly base, providing valuable information such as maximum and minimum temperatures. As well as providing the average of maximum, minimum temperatures, and average of rainfall amount on a specific month. This goal can be achieved by using pivot tables. Because “a pivot table is a highly flexible contingency table, that can be created from a large dataset and offers the possibility to look at one section at a time” (Helena & Agathe, 2012, p. 117). Due to easily navigate between each month and location, a timeline and slicer are employed.

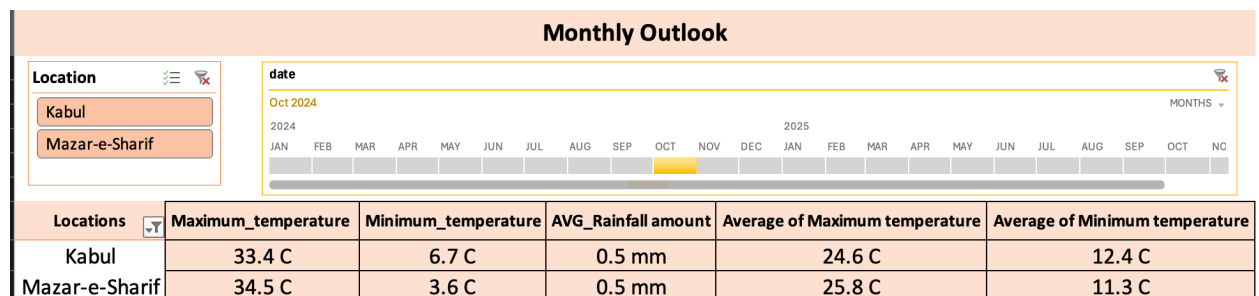


Figure 2: Monthly Outlook, Weather dashboard

Pivot chart (liner graph outlook):

“Pictures can be worth a thousand words” (Andrea, 2014, p. 176).

Therefore, the pivot chart (line chart) in this dashboard provides valuable information. For example, this liner graph contains the maximum and minimum temperatures in each month. Ad described the line graph is a visually comparable interface that shows the increase and decrease of temperature in different months.

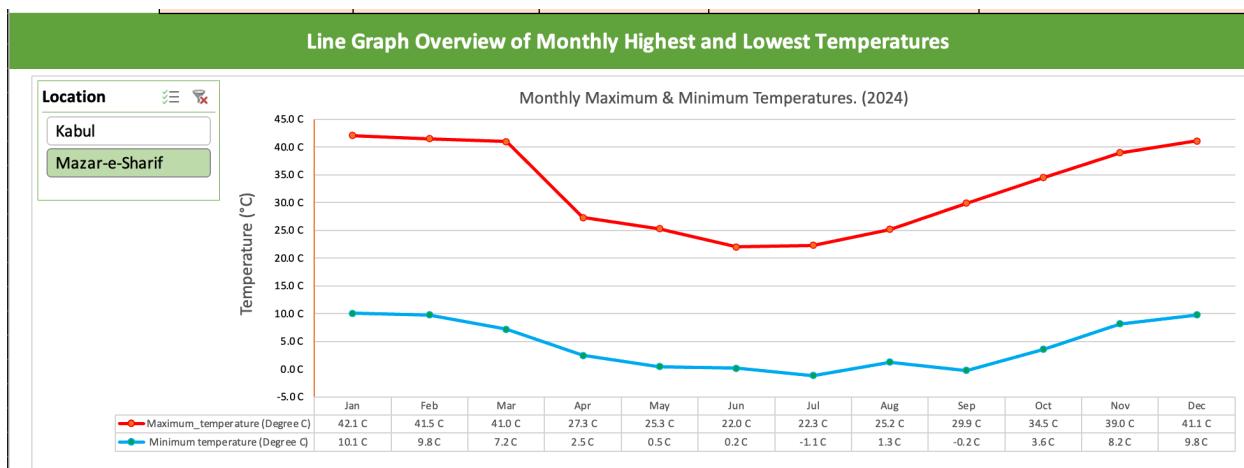


Figure 3: pivot Chart, Weather dashboard

## Challenges and limitation:

Developing the weather dashboard in excel came with several challenges and limitations. Some of the weather records were incomplete and required estimated values. Using complex formulas like (XLOOKUP, VLOOKUP and COUNTIF) with large data sets is time consuming and slows down performance. Also Pivot tables and slicers require careful structure to ensure correct answers. Additionally Including too much information made the dashboard cluttered, while too little detail reduced usability. In addition, including too much information made the dashboard cluttered, while too little detail reduced usability. Therefore, Conditional formatting, data validation, and optimized pivot table design were used to manage the challenges.

## Future Developments:

To increase the efficiency and capabilities of the weather dashboard for better decision-making and greater effectiveness, various features can be added to the dashboard in the future. The dashboard will connect to live weather data sources which will fetch the real-time weather condition and update automatically. Additionally, the development of the web version of the dashboard will make it more accessible. Automatic data updates, integrating real-time weather feeds and web base version could make the dashboard more effective in the future.

## Conclusion:

Development of the weather dashboard which demonstrates data analysis and uses Excel tools to track weather conditions on a daily and monthly outlook. By using the capabilities (slicer, pivot tables, condition formatting and data validation) the dashboard Provides an ideal and pleasant environment. According to Michael and John's (2013) dashboard designee principle, such as simplicity, clear labeling, and optimized formatting were applied to design the dashboard more effectiveness. Challenges like incomplete data, and balancing simplicity and detail were addressed through formulas, lookup functions, and pivot tables. Finally, the functionality and efficiency of this dashboard will be increased in the future by developing the web-based version of the dashboard and linking it to a live data source.



## References:

- Dierenfeld, H., & Merceron, A. (2012). *Learning analytics with Excel pivot tables* (pp. 115-121). [research.moodle.org](http://research.moodle.org).
- Miller, A. (2014). Application of Excel® Pivot Tables and Pivot Charts for Efficient Library Data Analysis and Illustration. *Journal of Library Administration*, 54(3), 169–186. <https://doi.org/10.1080/01930826.2014.915162>